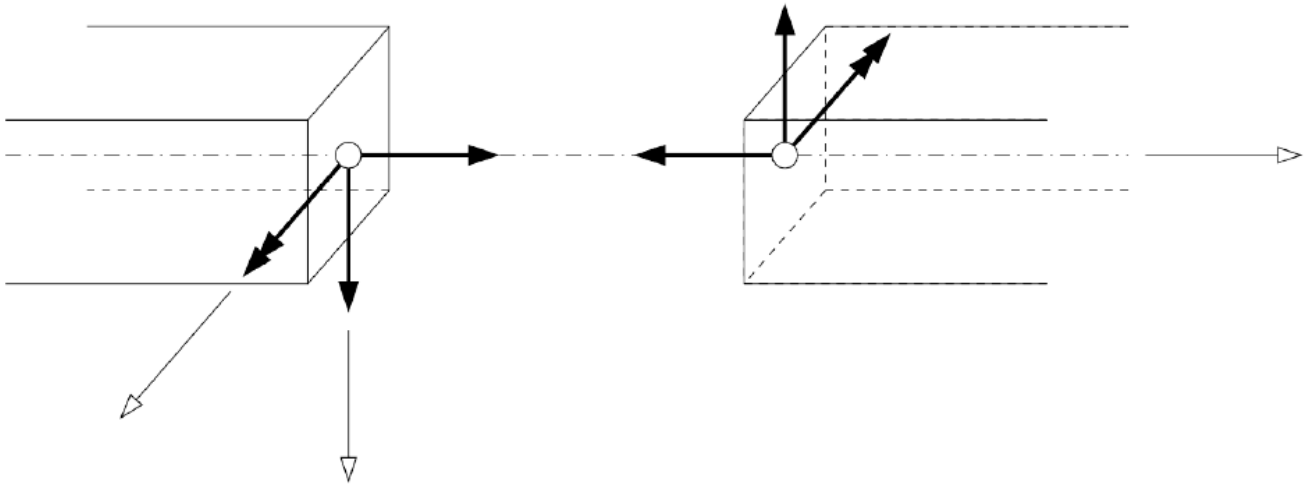
Engineering Mechanics I

Lecture



4. Fundamentals

Statics is the study of forces on bodies at rest.

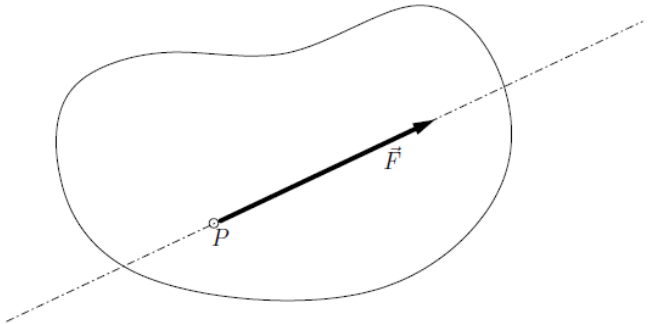
The Force

- Concept of force from daily experience: Example muscle force
- Forces cause a **change of the state of motion** and **deformations** of bodies, if not prevented by other forces.
- In a first step, real bodies are assumed to be rigid, they do not experience deformations.

4.1 The Force

A single force is uniquely defined by:

- its magnitude (amount)
- its direction (line of action)
- its point of application



4.1. The Force

- The unit of the physical quantity "force":

$$1[N] \text{ Newton} = 1kg \cdot 1\frac{m}{s^2}$$

- Amount and direction mathematically define a vector
- As vector symbol mostly $\vec{F}$, at drawing *F*, otherwise common **F**, F
- Force with fixed point of application is a **bound vector**
- Further types of vectors are line-volatile and free vectors, which all play a role in mechanics
- Vector calculus elementary for mechanics.

4.1. The Force

Elementary vector algebra – service

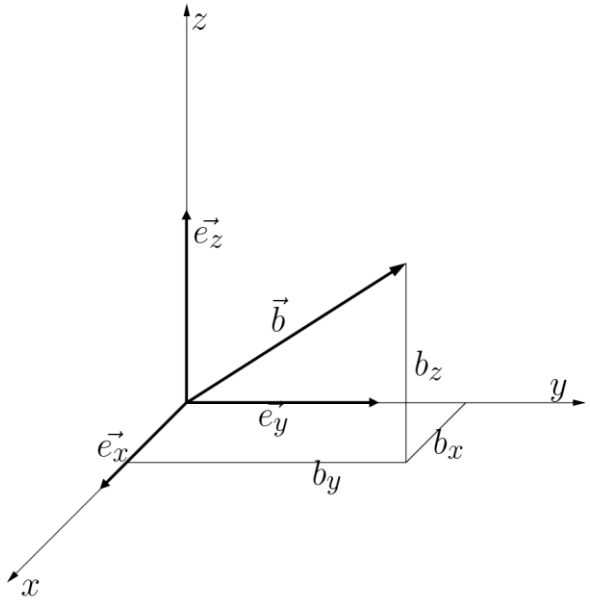
Component representation:

$$\vec{b} = b_x\vec{e}_x + b_y\vec{e}_y + b_z\vec{e}_z$$

x, y, z : Cartesian Coordinate System

$\vec{e}_x, \vec{e}_y, \vec{e}_z$: Orthonormal Vectors

$$\vec{b} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$: Component representation



4.1. The Force

Elementary vector algebra – service

Multiplication with a Scalar

$$\vec{b} = \lambda \vec{a}$$

$$\begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix} = \begin{pmatrix} \lambda a_x \\ \lambda a_y \\ \lambda a_z \end{pmatrix}$$

Vector Addition

$$\vec{c} = \vec{a} + \vec{b}$$

$$\begin{pmatrix} c_x \\ c_y \\ c_z \end{pmatrix} = \begin{pmatrix} a_x + b_x \\ a_y + b_y \\ a_z + b_z \end{pmatrix}$$

Magnitude of a vector

$$\left|\vec{b}\right| = b = \sqrt{\vec{b} \cdot \vec{b}} = \sqrt{b_x^2 + b_y^2 + b_z^2}$$

Scalar product

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\left(\measuredangle(\vec{a}, \vec{b})\right)$$

Orthogonality of two vectors

$$\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0$$

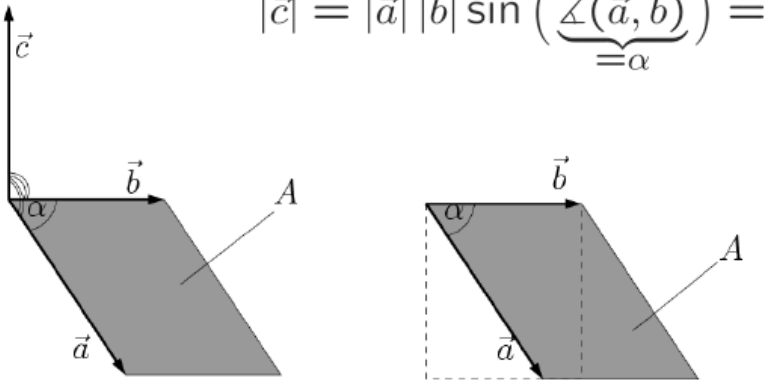
4.1 The Force

Elementary vector algebra – service

$$\vec{c} = \vec{a} \times \vec{b} = \begin{pmatrix} a_y b_z - b_y a_z \\ a_z b_x - b_z a_x \\ a_x b_y - b_x a_y \end{pmatrix}$$

$\vec{c} \perp \vec{a}$ and $\vec{c} \perp \vec{b}$

$$|\vec{c}| = |\vec{a}|\,|\vec{b}| \sin\left(\underbrace{\measuredangle(\vec{a},\vec{b})}_{=\alpha}\right) = A$$

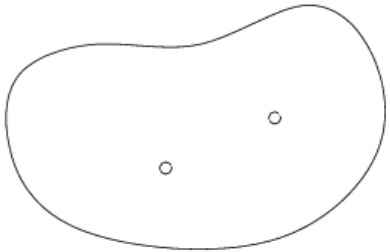


4.2 Axioms of statics

Greek: "axiom" = "to hold to be true".

- Fundamental theorem
- Obvious without proof
- Do not need or cannot be proven

4.2.1. Axiom of equilibrium (Newton)



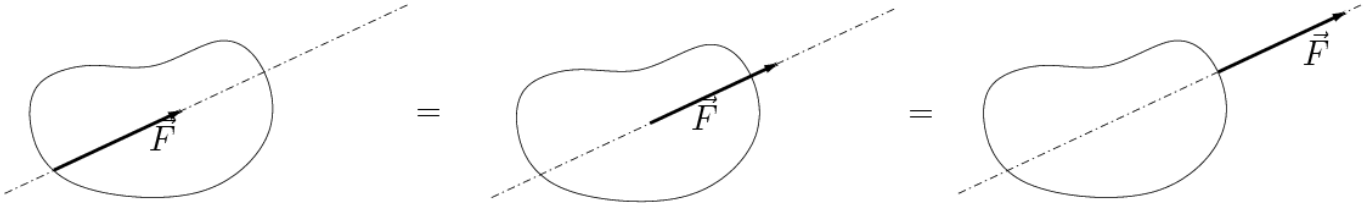
The vector sum of the forces of an equilibrium group results in 0.

Box 4.1

4.2.2. Axiom of Transmissibility (Varignon)

Statics of rigid bodies

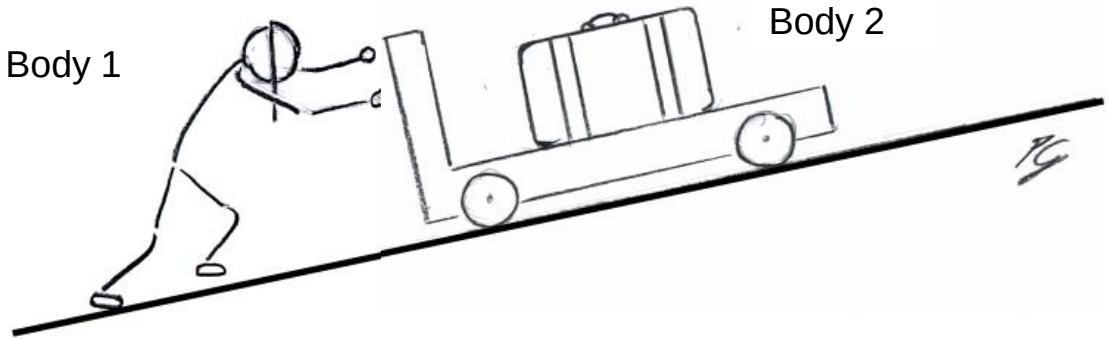
- Force can be shifted arbitrarily on a line of action
- Transmissible vector



- For deformable, elastic bodies no longer valid, point of action is decisive

4.2.3 Axiom of Interaction (Newton)

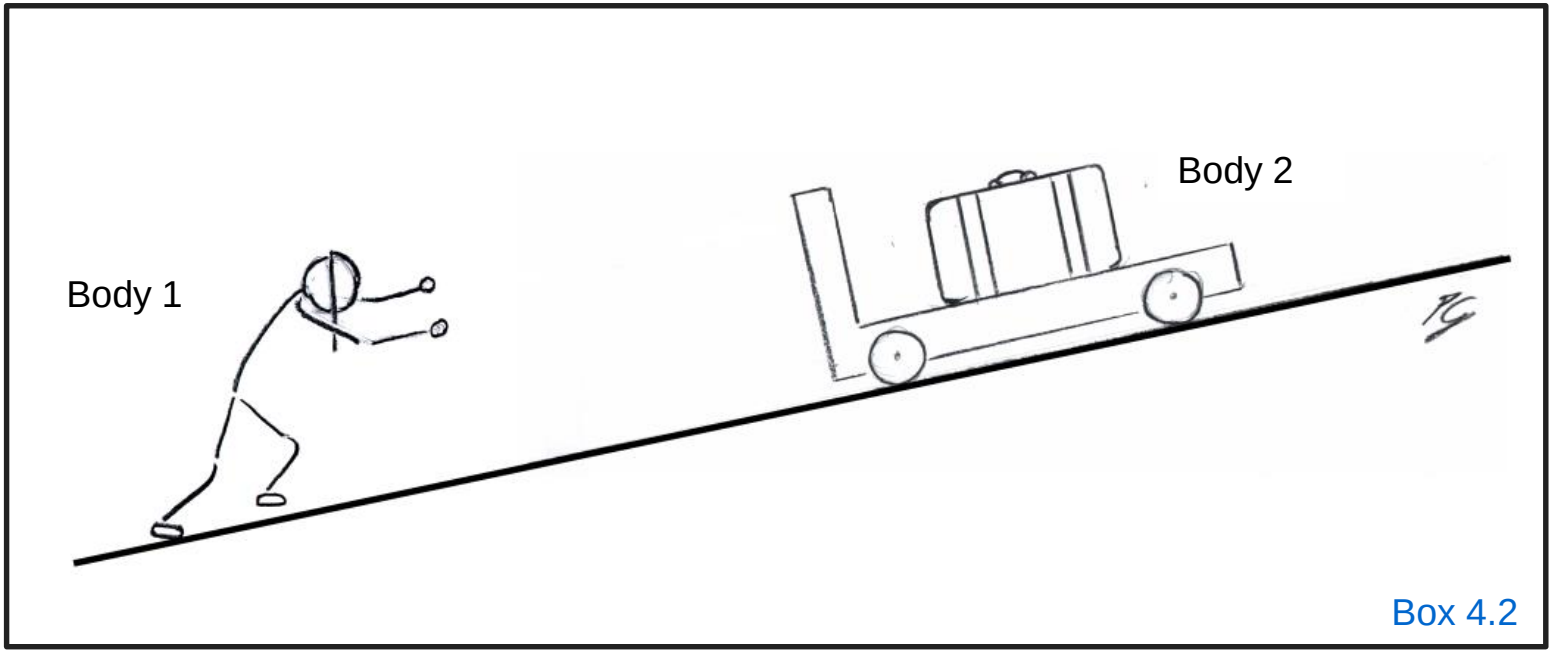
(Actio = Reactio, NEWTON)



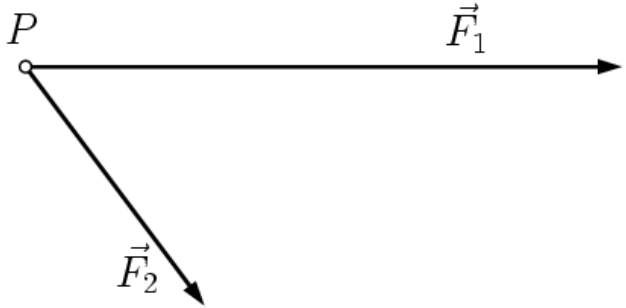
Box 4.2

4.2.3 Axiom of Interaction (Newton)

(Actio = Reactio, NEWTON)



4.2.4. The axiom of parallelogram of forces (Stevin)



Box 4.3

- F_1 and F_2 are statically equivalent to R
- Vector addition is commutative

4.3. Applications of the axioms

- Solving basic mechanics tasks

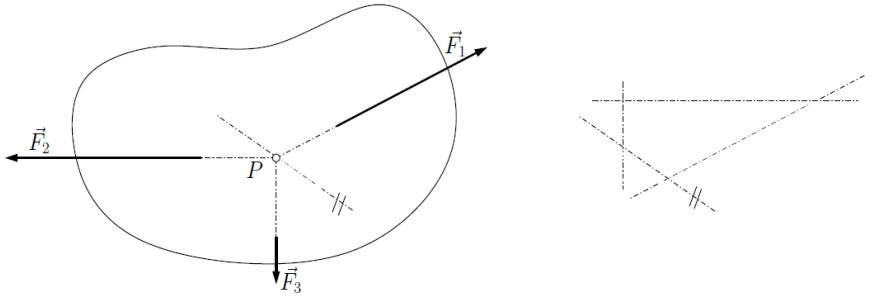
Assumption:

- Rigid bodies in a plane, on which
  - single forces act,
  - whose lines of action intersect in one point.

➢ Concurrent force system

4.3.1. Resultant of concurrent force system

- **Graphical reduction** to resulting force action with the same static effect.

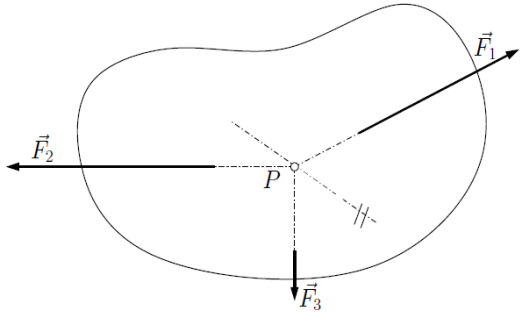


Box 4.4

- **Resultant** is vector sum of all attacking forces:

Box 4.5

4.3.2. Balance of forces with common point of attack

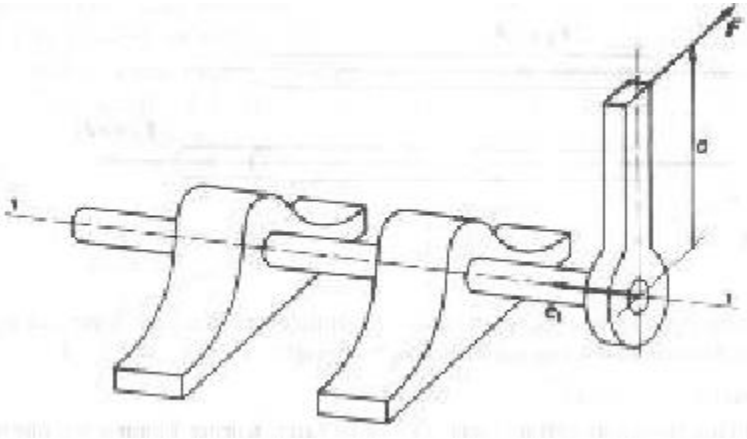


Box 4.6

Equilibrium condition:

Box 4.7

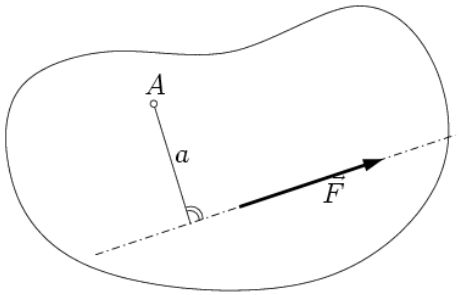
4.4. Moment





- No concurrent force system, lines of action do not intersect at one point
- Moment as a measure of the tendency to rotate

4.4.1. Moment of a Force



- Force $\vec{F}$ generated with respect to A:

$$M_A = aF\,[Nm] \text{ z.B. } \left(\circlearrowleft +\ -\circlearrowright\right).$$

- define positive direction of rotation
- distance *a* is named **lever arm**
- line volatility remains
- Moment of F at reference point
- bound

Resulting moment of multiple forces:

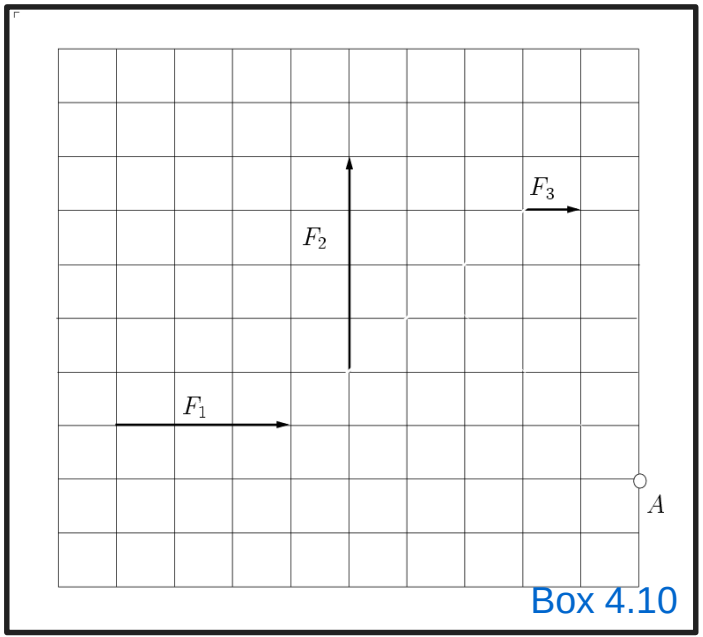
Box 4.8

4.4.1. Moment of a Force

Example 4.1: Numerical example for the moment

Required: Moment of forces F_1 to F_3 with respect to A.

Box 4.9



Box 4.10